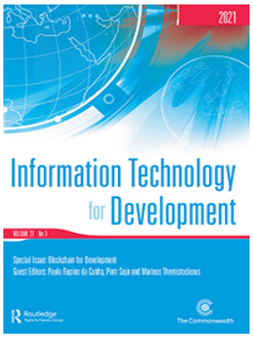

Information Technology for Development

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/titd20

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To cite this article: Jifan Ren, Jialiang Yang, Mengyang Zhu & Salman Majeed (2021): Relationship between consumer participation behaviors and consumer stickiness on mobile short video social platform under the development of ICT: based on value co-creation theory perspective, Information Technology for Development, DOI: 10.1080/02681102.2021.1933882

To link to this article: https://doi.org/10.1080/02681102.2021.1933882

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INFORMATION TECHNOLOGY FOR DEVELOPMENT
https://doi.org/10.1080/02681102.2021.1933882

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Relationship between consumer participation behaviors and consumer stickiness on mobile short video social platform under the development of ICT: based on value co-creation theory perspective

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ABSTRACT
As a representative of the sharing economy business model, TikTok, a short-form mobile video-sharing social networking platform, has numerous consumers worldwide. Based on the background of the development of information technology and value co-creation theory, this paper analyzes the relevance between information and communication technology and the sharing economy; studies the internal mechanism between consumer participation behaviors and consumer stickiness under different emotions; and expounds the impact of consumer participation behaviors on consumer perceived value. The structural equation model was used to analyze 902 questionnaires collected from TikTok consumers. The research results reveal that positive consumer participation behavior has a positive effect on consumer perceived value, negative consumer participation behavior harms consumer perceived value, and consumer perceived value has a positive impact on consumer happiness and stickiness. Consumer happiness plays an intermediary role between consumer perceived value and consumer stickiness.

KEYWORDS
Sharing economy; ICT; value co-creation; consumer participation behaviors; consumer perceived value; consumer stickiness

1. Introduction

The sharing economy is the peer-to-peer sharing of underutilized goods, services, and other resources that prioritizes right to use above ownership (Schor & Fitzmaurice, 2014). With the widespread use of the internet, internet of things, big data, cloud computing, artificial intelligence, information communication technologies (ICT), and the separation of right to use and ownership (Puschmann & Alt, 2016), the growing focus on personal sharing (Bardhi & Eckhardt, 2012) and relying on platforms (Belk, 2014), the sharing economy has been broadly adopted in many industries and has achieved success in the innovation of business models (Martin, 2016), such as Uber, Mobike, and Airbnb.

In recent years, sharing economy enterprises have exhibited exceptional vitality and become a new impetus for the world's economic growth with advancing information technology (IT) playing an important role in its rapid development. Scholars agree that IT is a key factor in

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Xusen Cheng is the accepting Associate Editor for this manuscript.
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socioeconomic and ICT development, which has engendered digital opportunities for many countries and industries (Alderete, 2017). ICT-based networks are part of a broader global movement; ICT-based information processing and transmission have innovated across various spectrums of the interconnected society (Mclennan, 2016), including the sharing economy. Information communication technology (ICT) has fueled the contribution of the sharing economy enterprises to the world's economic growth. Extant literature on ICT and the sharing economy mostly focus on specific industries or demographic features (e.g. Margariti & Travlou, 2018). This study systematically analyzes the role of ICT in the development of the sharing economy which is yet under-explored. TikTok, a platform for mobile short video sharing (Hu & Guo, 2020), has fueled the importance of ICT in the sharing economy with consumers' participation, interaction, and attachment.

ICT-based communication networks engender digital opportunities across the globe for interconnected societies (Alderete, 2017; Mclennan, 2016), including the sharing economy. Digitalization in the fields of big data platforms drives the sharing economy, which involves a reciprocal exchange of tangible and intangible resources, such as information (Felländer et al., 2015), to realize the advantages of peer-to-peer communication, i.e. person-person communication, not company-person.

The concepts of sharing economy and value co-creation are interlinked. Value refers to the sum of tangible or intangible benefits for consumers, while value co-creation refers to the process of creating goods and services of value (Kotler & Keller, 2012). A consumer is noted as the ultimate value creator under the sharing economy business model. This supports the notion that consumers' interaction with each other ignites value co-creation in the sharing economy (Christian, 2009). The sharing economy creates values for the participants (Gobble, 2017) and is inseparable from collaboration and interaction (Kumar et al., 2018). Prahalad and Ramaswamy (2004) note that 'unique values with consumers' participation', i.e. value co-creation, may resolve the dilemma faced by the traditional production-sale system in the twenty-first-century. The sharing economy is considered as an advanced value co-creation process among sharing enterprises to promote and coordinate interdependent economic actors through platform participation (Breidbach & Brodie, 2017). Although some research analyzed the sharing economy based on the value co-creation theory (Wang et al., 2019; Zhang et al., 2018), consumers' participation in the ICT-based communication platforms to co-create valuable production and consumption experiences is widely unexplored in the extant literature on the sharing economy.

Short-form mobile videos are video applications with functions of short video shooting, editing, uploading, and sharing. The scale of consumers of short-form mobile video social platforms has rapidly expanded worldwide and has ranked third globally in the number of consumers of social software. With the 'mobility' of people's lifestyles, short-form mobile videos have developed into mainstream formats due to its wide coverage of theme content, creating an industrial value chain and becoming a new outlet for content entrepreneurs and the breakthrough point of a new round of economic growth (Hu & Guo, 2020). Given the instrumental role of ICT in economic growth, the importance of short-form mobile video platforms in the development of the sharing economy is not explored yet.

As a representative of the sharing economy social platform, TikTok has contributed to unprecedented economic growth across the globe and optimized this as a development opportunity with consumers' participation, innovation, and interaction behaviors (Wang et al., 2019). To achieve the study objectives, we conducted an online survey to gather data from the Chinese users of TikTok, an online platform for short-form mobile video. This study attempts to bridge the research gaps with the analysis of value co-creation and ICT to understand consumers' behaviors in the sharing economy. We use TikTok as the research object for the following: to study the internal mechanism between consumer participation behaviors and consumer stickiness under different emotions, to study the impact of different consumer participation behaviors on consumer perceived value, to further augment the research on consumer value co-creation behavior, and to expand and enrich the value co-creation theory under a sharing economy model.

2. Literature review

2.1. ICT and the sharing economy

ICT has become one of the important industrial sectors that boosts the production and employment levels in parallel to contributing human progress (Lee et al., 2017) and the world economic development (Heo & Lee, 2019). Consequently, governments, international donors, and non-government organizations have started to invest heavily in global ICT initiatives to improve the living conditions of various communities (Singh et al., 2018). Thus, ICT has attracted the interest of scholars across fields, such as finance (Chatterjee, 2020), agriculture (Birke & Knierim, 2020), trade (Molony, 2009), hospitality and tourism (Majeed et al., 2020a)

The sharing economy is a relatively new economic model that has developed rapidly in the past decade, bringing together distributed networks of individuals to share or exchange otherwise underutilized assets (Koopman et al., 2015). Participants of the sharing economy rely on collaborative consumption by sharing access to products or services, such as a bedroom, office space, or a car seat, rather than emphasizing individual ownership and coordinating those experiences entirely online (Cheng et al., 2019).

The sharing economy is closely related to ICT. Technological developments have simplified the sharing of physical and non-physical goods/services through the availability of various information systems on the Internet, thus giving birth to the sharing economy (Hamari et al., 2016). The sharing economy's development is primarily influenced by the increasing penetration of information technology (IT) (Frey et al., 2017), e.g. the Internet, smartphones, social media, etc., and the creation of marketing platforms (Felländer et al., 2015).

From the perspective of value creation, IT contributes to the matching of supply and demand as an operant resource. IT is considered important in improving the transparency of value creation activities, thus building trust among participants (Lusch & Nambisan, 2015). For example, TikTok uses IT as an operant resource to directly contact consumers and support interactions between participants, such as collecting direct feedback from consumers, promoting complaint management, and proactively identifying users with potential problems and taking corresponding management measures. This phenomenon improves consumer happiness, consumer perceived value, and consumer stickiness. Therefore, the development of the sharing economy is linked to the development of ICT.

According to Eckhardt et al. (2019), the sharing economy entails economically motivated access (Eckhardt & Fleura, 2016) rather than socially motivated sharing (Belk, 2010). Moreover, in the sharing economy, prosumers may be both a producer and a consumer. TikTok is a social media, but its characteristics that distinguish it from traditional social media are its economic attributes which are the typical characteristics of the sharing economy. TikTok consumers often share videos out of economic motives. Consumers' interactive behavior likely enables them to watch, forward, share, and evaluate their favorite content on TikTok. TikTok consumers are both video producers and video consumers. The video producers have the willingness to share the excess value in the form of videos, thus, realizing the interaction between the needs of producers and consumers of videos. Consumers' interactive behavior creates value for video producers, and economic value is the direct manifestation of using TikTok.

2.2. Value co-creation in the sharing economy

The sharing economy, due to its 'sharing' nature, determines the emergence of value co-creation phenomenon among its different participants. Compared with the dominant traditional ownership market model, the concept of temporary use and sharing of resources attracts consumers owing to the high convenience and lower price (Puschmann & Alt, 2016).

Consumers' participation to create unique values, i.e. value co-creation, may resolve the complexities of resource sharing and utilization in the traditional production-sale system (Majeed et al.,

2020b; Prahalad & Ramaswamy, 2004). This concept laid the foundation of the 'experiential value theory' (Prahalad & Ramaswamy, 2004), 'service-oriented logic theory' (Vargo & Lusch, 2004), 'co-creation process theory' (Payne et al., 2008), 'co-creator role theory' (Grönroos, 2008), and 'co-creation behavior theory' (Yi & Gong, 2013). These influential schools have conducted pioneering researches on the experience value, leading logic, internal mechanism, role, and behavior of participants of value co-creation, making value co-creation a significant theoretical system in marketing management.

Wang et al. (2019) also stated that the essence of the sharing economy is to rely on the platform to enable users to achieve value co-creation. Through value co-creation, competitive advantage is achieved (Ballantyne & Varey, 2006). Under the traditional economic model, according to the supply and demand of services (Navarro et al., 2014), the role of value creation splits into the activities of providers (enterprises) and consumers (consumers) (Grönroos & Voima, 2013; Payne et al., 2008). However, under the new resource allocation model of the sharing economy, the resources required for services come not only from enterprises but also from idle resources owned by individuals. Individuals have replaced enterprises as the main providers of resources, whereas enterprises have assumed roles behind the scenes and become platforms to promote service supply and demand connections (Ramaswamy & Ozcan, 2018). The dominance under the Service-Dominant logic starts to shift from the provider to the consumer, when the Consumer-Dominant logic emerges; the consumer completely becomes a leader. The Consumer-Dominant logic relocates value-creating agency into consumers' living environment and goes beyond specific visible service interactions that are normally in the sphere of service providers' influence, rendering consumers' value-forming processes 'invisible' to service organisations (Medberg & Heinonen, 2014).

The sharing economy is an economic system in which agency enterprises utilize online platforms to build a peer-to-peer network that can generate trust and reduce transaction costs for consumers (Ranjbari et al., 2018). Online platforms assist enterprises to access resources and consumers with idle resources to facilitate transactions without transferring ownership to consumers. Therefore, value co-creation in the sharing economy exists between consumers on both sides and not between enterprises and consumers, such as in the traditional economy. Thus, the sharing economy has reconstructed the way in which value is created in the traditional economy (Yang Xuecheng & Ke, 2016).

TikTok is also a kind of sharing economy (see Figure 1). Besides entertainment, short-form mobile videos have other important uses, for example, the current transmission of traditional culture on

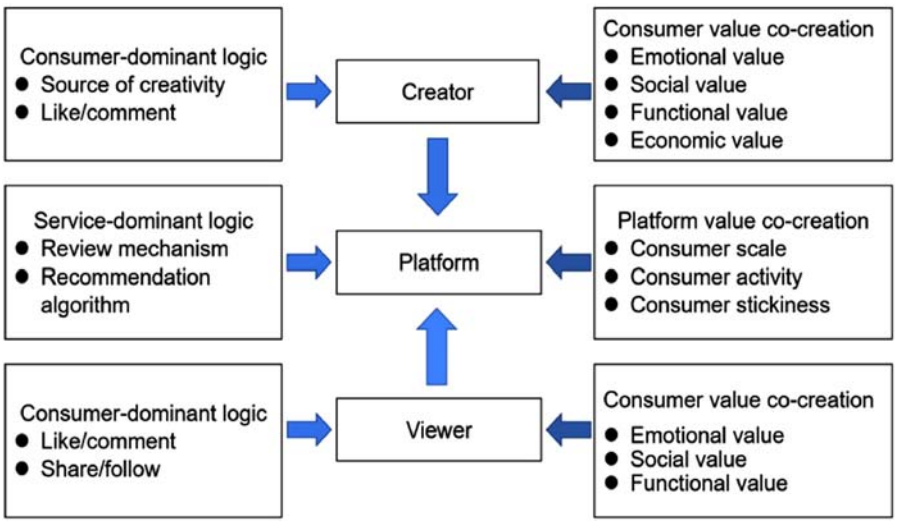

Figure 1. The value co-creation process of TikTok.

these platforms presents the characteristics of high efficiency, visualization of communication effect, and wide coverage of theme content (Hu & Guo, 2020). The huge consumer group of TikTok can be divided into two categories, namely video creators and video viewers. Regarding short videos as products and sharing assets, the creators are the producers, the viewers are the consumers, and TikTok, as an enterprise, realizes the value co-creation in product design, production, and distribution.

In the design process, the TikTok video creator can determine the video theme and content based on the number of likes, views of other videos, comments from viewers, and their own creative sources. At certain moments, viewers will also be inspired to create videos, and thus turn into creators. TikTok participates in the review and distribution of videos. Only the videos that fulfill the audit conditions can be uploaded, and the distribution of videos is part of the production process that is determined by the algorithm concerning a single user's browsing and like record (that is, the video preference). The process encompasses the platform creating value together with creators and viewers, which enables the platform to gain a larger consumer scale and enhance consumer activity and stickiness.

During the video playback, the comments of ordinary viewers increase the interest in the video and strengthen the interaction among viewers and between creators and viewers. This also enhances the fan stickiness between creators and viewers. Gao et al. (2020) stated that content creators on TikTok act as consumers and replace enterprises as the main providers of resources, and enterprises operate behind the scenes and become platforms to promote service supply and demand connections.

2.3. Consumer perceived value

> "Consumer value" is the difference between total consumer value and total consumer cost for the consumer about a product/service (Kotler, 2012). When consumers perceive that a product/service has value, they think that consuming the product/service is worthwhile, at which point the consumer secures the corresponding consumer perceived value (Zeithaml, 1988).

The consumption value theory divided consumer perceived value into five types of value: functional, social, emotional, cognitive, and conditional (Sheth et al., 1991). Some scholars proposed a three-dimensional value structure, i.e. functional, social, and emotional, of consumer perceived value (Roig et al., 2006; Sanchez et al., 2006). In this research, we measure consumer perceived value in four dimensions: emotional, functional, social, and economic value (Zhang et al., 2017).

Emotional value is the value perceived due to the satisfaction of emotional needs (Zhang et al., 2018). Functional value encompasses the positive benefits perceived from product/service characteristics satisfying individual needs (Sheth et al., 1991). Social value is the value perceived when social relationships or like-minded partners are found when purchases are made (Zhang et al., 2018). Furthermore, economic value refers to the economic revenue perceived when using goods/services.

At short-form mobile video social platforms, social and emotional values tend to have a significant impact on consumers' purchase behavior (Majeed et al., 2017; Martin, 2016). Moreover, short video consumer inspiration is the embodiment of the consumer inspiration concept, which reveals the trigger mechanism for consumer inspiration in marketing and its impact on consumer integration. It helps anchors and platforms to produce more efficient marketing content and strategies. It also helps consumers to improve the consumption experience of short video shopping (Gao et al., 2020). Thus, on short-form mobile video social platforms, it is necessary to satisfy the value needs of consumers (Zeithaml, 1988).

2.4. Consumer stickiness

In the study of online consumption behavior, researchers found that the conversion cost of the network is very low and retaining consumers is not an easy task. Therefore, scholars began paying attention to the stickiness of online network behavior, explored how to strengthen the

consumer's dependence on the website from the perspective of website design, and proposed the concept of website stickiness (Li et al., 2006; Lin, 2007). With the development of the internet, consumer stickiness has been widely used in across various disciplines of online research.

Beddoe-Stephens (1999) defines consumer stickiness as the ability of a website to continuously encourage consumers to browse repeatedly. Consumer stickiness is a term specifically used to describe the ability of a website to attract consumers to stay on the webpage for a longer duration, while it also links stickiness with the consumer's repeated willingness to purchase (Zott et al., 2000).

Consumer's willingness and behavior of repeated purchases or long-term use are primarily caused by the fact that consumers gradually became the dominant part of the market, and scholars began shifting their research accordingly. For example, Lin et al. (2010) defined stickiness as the consumers' time that is spent on a company's social network. Some scholars explain consumer stickiness as all the costs that consumers need to pay when they make changes in consumption behaviors or habits. Wang (2009) believes that stickiness means consumers need to pay a lot of extra costs when considering changing their consumption behaviors or habits. To avoid such losses, consumers consciously maintain their original habits and behaviors.

2.5. Consumer happiness

The primary focus of consumer behavior research has shifted from satisfying consumer needs to enhancing consumer happiness (Gong & Yi, 2018). Sksjrvi et al. (2017) stated that consumer happiness can affect consumers' current or future buying decisions. Therefore, consumer happiness (or subjective well-being, SWB) is of research value.

SWB refers to an individual's cognitive and emotional evaluation of his/her overall quality of life (Diener, 2000), which includes the cognition of life satisfaction and emotional experience of events. Emotional experience includes two aspects: positive emotion (happiness, satisfaction, etc.) and negative emotion (depression, low self-esteem, etc.). Similar to the general research on SWB, at first, the research on online social consumers' SWB used the measurement indicators of general SWB (e.g. Chiu et al., 2013; Manago et al., 2012).

In the study of the relationship between online social use and SWB of consumers, we found that demographic factors can affect the relationship. For example, the study by Seder and Oishi (2009) investigated the relationship between the racial nature of Facebook friends and the SWB of consumers. The results revealed that for European and American nationality consumers, there is a strong positive correlation between the racial nature of Facebook friends and the SWB of consumers, whereas, for non-European and non-American nationalities, the racial nature of friends is not significantly related to their SWB. Online social circles have become an important variable in the study of the relationship between online social and consumers' SWB (Nabi et al., 2013). Research shows that if an individual feels close or connected with others, his/her happiness level is higher (Reis et al., 2000), that is, the larger his/her online social circle, the higher his/her subjective happiness (Kim & Lee, 2011).

3. Development of hypotheses

3.1. Relationship between consumer participation behavior and consumer perceived value

Consumers are active players to co-create experiences (Vargo & Lusch, 2016) and need to be present, get involved, and interact with people, products, and services (Prebensen et al., 2013). Consumer participation refers to the extent to which consumers contribute effort, preferences, knowledge, or other inputs to the various service process stages including service development, production, delivery, and recovery (Dong & Sivakumar, 2017). It consists of three major forms of participation: attitudinal, information, and actionable (Chen & Raab, 2016). Attitudinal participation is the

willingness to contribute to the value creation process (Jiang et al., 2019). Information participation comprises informational contributions to the service process (Hsiao et al., 2015). Actionable participation represents behavioral contributions to the service process.

With the intensified competition of short-form mobile video-sharing social platforms, consumers' participation behaviors are more driven by value; empathy may also trigger consumer participation, like the example of cancer survivor groups (Carrier et al., 2015). Viewers on TikTok can gain functional and emotional value when watching videos; simultaneously, under the influence of empathy, some viewers will also have creative desires and try to create short videos that may further bring value to other consumers.

From the perspective of consumer willingness, consumer participation behaviors can be divided into positive and negative behaviors. A consumer's interest and love for the particular content of a video is shown in terms of the time he/she consumes to watch the video. Consumers' willingness to spend time and energy to watch the content of a video and share it with others satisfies their curiosity and social needs of belonging with followers, appreciation from others, and staying connected with the society. Besides the oft-quoted purpose of entertainment (Hu & Guo, 2020), TikTok attracted the attention of religious scholars, educational experts, destination marketing organizations (DMOs), and politicians, to name a few, to disseminate their messages across its millions of users.

The social value and the connotation of value co-creation behind short-form mobile video-sharing platform alongside its conventional functions of video watching and dissemination of information are the reasons for its popularity for consumers' stickiness which differentiates it from other social media, such as Facebook, twitter, and Instagram. TikTok consumers participate in the content innovation by sharing their knowledge, information, suggestions, experiences, and new ideas to develop new products/services or to simply improve the existing products/services. Consumers' interactive behaviors at short-form mobile-video platforms, such a writing comments and forwarding videos, show their active participation in the content creation and innovation. Consumers create their own short-form videos to attract the attention of other unfamiliar consumers and appreciate each other with likes and comments to show their love for video creators.

When consumers have more sense of control and experience a sense of independence in the service process, they choose to actively participate, which will not only serve at the stage of money as an incentive but also satisfy the psychological needs to obtain the consumer's perceived value (Bateson, 1985). Schneider and Bowen (1995) pointed out that many items, such as improving efficiency, reducing price, improving the sense of control, having freedom and opportunity, reducing waiting time, and enjoying greater customization, can motivate cooperative production. Users often need to put in various levels of effort in the firm-initiated interaction activities in social media, and this could affect the perception of value (Quach et al., 2020). Given the unprecedented popularity of short-form mobile video platform,

Conversely, when consumers passively participate in the cooperative production process, their sense of control over the process will be reduced, and they will not be able to effectively reduce risks, control results, and obtain psychological satisfaction through participation (Lloyd & Luk, 2011); In the process of video content dissemination, there will be situations of imitating creation, leading to the appearance of a large number of similar content, consumers will not be able to obtain the pleasure of service, improve service satisfaction, reduce anxiety and feel social comfort, and gain other internal benefits (Markley & Davis, 2006), which means that a negative impact on consumer perceived value will be the result. Jung and Yoo (2017) also said that positive CCI (consumer-to-consumer interactions) and dysfunctional consumer behavior (negative CCI) influence consumers' affection for service.

Therefore, this paper proposes the following hypotheses:

H1: Positive consumer participation behavior has a positive impact on consumer perceived value.

H2: Negative consumer participation behavior has a negative impact on consumer perceived value.

3.2. Relationships among consumer perceived value, consumer happiness, and consumer stickiness

Hedonism theory identifies happiness as pleasure (Haybron, 2008). Niedermeier et al. (2018) defined happiness as a specific form that involves consumer satisfaction, consumer delight (joy, pleasure, surprise, etc.), and consumer harmony (involves emotional harmony such as feeling calm, relaxed, and cognitive harmony such as peace of mind, being unconcerned).

When using TikTok, consumers attain perceived value: by recommending short videos that meet the interests and needs of viewers, TikTok enables viewers to gain pleasure and knowledge, thereby realizing their emotional and functional value. Creators gain likes, comments, and an increase in the number of fans through the videos posted, thus realizing their social and emotional value. Some become Internet celebrities, who can start to sell products and advertise to obtain income and economic value. Both viewers and creators can perceive social value.

This value co-creation process enlarges consumers' social circle at the same time. As Kim and Lee (2011) said, the larger a user's online social circle, the higher his/her subjective happiness. The positive benefits that consumers perceive from products/services may make an individual's online social circle larger or the individual feel intimate or connected with others, thereby improving consumer happiness.

Therefore, this paper proposes the following:

H3: Consumer perceived value has a positive impact on consumer happiness.

For short-form mobile video-sharing platforms, how to retain consumers is a crucial issue. Therefore, they need to increase consumer stickiness. What factors affect consumer stickiness? From the user's perspective, some scholars have begun to focus on their perceived value and found that there is a connection between the two.

A plethora of literature presents that consumers' perceptions may impact their behaviors (Majeed et al., 2018; Xue et al., 2020). Through the empirical analysis of social networks, Fang and Fang (2016) verified that perceived entertainment can affect consumer stickiness; Ye (2015) also verified that consumer perceived value affects consumer stickiness through consumer satisfaction when researching on consumer satisfaction of the online Q & A community; Liu and Zhang (2016), based on the 'perception-emotion-willingness' model, found in their empirical study that functional and emotional value positively affect consumers' repeated purchase willingness by affecting their psychological emotions, which acts as an intermediary variable in the model; El-Adly and Eid (2016) found that consumer perceived value has a significant positive effect on consumer satisfaction and loyalty to malls; Chen et al. (2019) also found that information quality, perceived value, and community identification have positive effects on consumers' stickiness to online brand communities; Yuan (2018) conducted an empirical analysis on consumers' continuous use willingness of the mobile reading platform and found that perceived usefulness, ease of use, and entertainment have a significant positive impact on consumer satisfaction and stickiness.

Based on the relationship between consumer perceived value and consumer stickiness in the above literature, we propose the following:

H4: Consumer perceived value has a positive impact on consumer stickiness.

Consumer happiness is conceptualized as consumers' perception of the extent to which their well-being and quality of life are enhanced (Gong & Yi, 2018). After many years of research, Oliver (1980) and other scholars have proved that there is a significant positive correlation between consumer happiness and consumers' willingness to purchase from both theoretical and empirical aspects. The research by Ha et al. (2010) showed that in an online transaction, the past positive experience of consumers will increase consumer satisfaction, and satisfaction will also promote their 'continuous use willingness' and loyalty. According to the theory of reasoned action model (Fishbein & Ajzen, 1975), behavior attitude influences behavior willingness, and continuous use willingness is behavior

willingness, while consumer happiness and overall satisfaction can be regarded as an attitude. A recent finding has highlighted that consumer satisfaction is positively associated with consumer stickiness (Keiningham et al., 2014). Therefore, consumer happiness will have a positive impact on consumer stickiness and a mediating effect between consumer perceived value and consumer stickiness.

Therefore, this paper proposes the following:

H5: Consumer happiness has a positive impact on consumer stickiness.

H6: Consumer happiness plays a mediating role between consumer perceived value and consumer stickiness.

3.3. Usage time, consumer participation behaviors, and consumer perceived value

'Usage time' refers to the length of time a consumer uses for short-form mobile video-sharing at social networking platform. To some extent, the time and frequency of consumers using short videos have a positive impact on their behavior willingness; the higher the level of participation, the stronger the behavior willingness (Ewing et al., 2020). Therefore, the longer consumers use short videos, the stronger the impact of their participation behaviors on perceived value.

From the perspective of preference and utility theory, the usage time will also affect consumer perceived value. All human activities are based on the motivation of utility maximization, and the extended utility function depends on the goods/services consumed in the current period and the stock of personal capital and social capital at that point in time (Becker, 2013). Personal capital refers to the consumption experience that can affect the present or future utility perception in personal experience, that is, when consumers make decisions, they consider the impact of the expected results on their capital and expected earnings and finally consider whether to make this choice. The preference and utility theory assumes that people consume energy, time, and goods to produce 'imagination' capital to better predict future utility and maximize the discounted value of present and future utilities (Becker & Mulligan, 1997). Therefore, the longer the consumer uses TikTok, the greater the time cost, so the consumer will care more about the effort he/she makes; thus, consumers' evaluation of the results tends to be positive or negative (Narasimhan, 2011). Therefore, as the usage time increases, the impact of both the positive and negative consumer participation behaviors on the consumer perceived value will be amplified.

Therefore, this paper proposes the following:

H7: Usage time has a moderating effect on the relationship between positive consumer participation behavior and consumer perceived value.

H8: Usage time has a moderating effect on the relationship between negative consumer participation behavior and consumer perceived value.

There is a dearth of literature that examines consumers' participation in the production and consumption of valuable experiences at online social media platforms in the context of the sharing economy. This study fills this gap by synthesizing different theoretical underpinnings, such as value co-creation theory, to unravel the notion of consumers' valuable production and consumption experiences gleaned with their participation in online shared communication platforms and used TikTok as an online platform for the study context. Thus, the importance of ICT in the development of the sharing economy is examined in this study which was previously under-explored.

Based on the value co-creation theory and the hypotheses above, this paper establishes a research model in Figure 2.

4. Method

4.1. Research object and data collection

This paper selected TikTok as it has a huge influence on the short-form mobile video-sharing industry, with over 820 million consumers and 10 billion market capital. This paper investigates

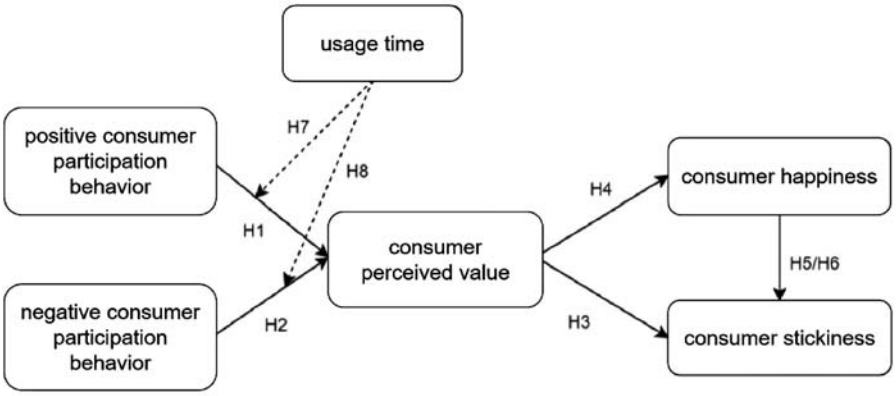

Figure 2. Resarch model.

TikTok consumers of different ages and usage times. It is hoped that through in-depth investigation, we can understand the internal mechanism of consumer value co-creation behavior and consumer stickiness on short-form mobile video-sharing social platforms derived from the development of ICT. A questionnaire survey through Wenjuanxing (an online data collection platform) aimed at TikTok consumers was conducted. About 1000 questionnaires were distributed, and 908 valid questionnaires were collected, making a response rate of 90.8%. All the participants used a random allocation method, so confounding variables and irrelevant factors were effectively controlled.

According to the data collected, the number of males is 484 (53.3%), and the number of females is 424 (46.7%). The respondents were aged 18–30 years (43.3%) and 30–40 years (37.8%); there were also a few people in other age groups. The educational level was mainly college or bachelor's degree (68.9%), and the usage times were generally 3–6 months (30.8%) and 6–12 months (33.8%), as presented in Table 1.

4.2. Measurement

The design of the questionnaire was based on the literature scale (Table 2), but the wordings have been changed slightly to adapt to the current context. Supported by the previous research literature, the questionnaire design was in line with the actual needs and characteristics of the research object, with all the items measured on a 7-point Likert scale (from 'strongly disagree' to 'strongly agree').

Table 1. Demographic characteristics ($n = 908$).

Personal information	Category	Number of cases	Percentage
Gender	Male	484	53.3%
	Female	424	46.7%
Age	Under 18 years	28	3.1%
	18–30 years	393	43.3%
	30–40 years	343	37.8%
	Over 40 years	144	15.9%
Educational level	High school and under high school	211	23.2%
	College or bachelor	626	68.9%
	Master and above	71	7.8%
Usage time	1–3 months	190	20.9%
	3–6 months	280	30.8%
	6–12 months	307	33.8%
	More than 12 months	131	14.4%

Table 2. Measurement items.

Construct	Items	Sources
Positive consumer participation behavior	I often watch/repost popular short videos of interest on a short-form mobile video-sharing platforms	Dai and Gu (2017)
	I often give likes/comment on short videos posted by others on a short-form mobile video-sharing	
	I often reply to comments/messages from others on short videos I publish on short-form mobile video social platforms	
	I often publish my short videos or participate in activities on short-form video social platforms	
Negative consumer participation behavior	I am not willing to participate in the activities published on short-form video-sharing social platforms	Järvi et al. (2018)
	I don't reply to comments/messages from others on short-form video-sharing social platforms	
	I don't actively create and publish short videos	
	I don't buy goods/services recommended on short-form video-sharing social platforms	
Consumer perceived value	I feel myself (my work) being concerned/recognized when my friends/fans interact with me	Dai and Gu (2017)
	I believe I can learn knowledge/information by using short-form video-sharing social platforms	
	I believe I can meet friends/gain fans by using short-form video-sharing social platforms	
	I believe I can get high cost-effective goods/services by using short-form video-sharing social platforms	
Consumer happiness	Short-form video-sharing social platforms inspire my creative inspiration and enthusiasm	Hsieh et al. (2018)
	I feel pleased when I watch popular videos on short-form video-sharing social platforms	
	I enjoy browsing short videos and interacting with friends or fans	
	Realizing my value in the process of using short-form video-sharing social platforms	
	Learning and mastering important knowledge/skills in the process of using short-form video-sharing social platforms	
	Using short-form video-sharing social platforms is helpful for my social relationships	
consumer stickiness	I will spend a long-time browsing videos on short-form video-sharing social platforms	Dai and Gu (2017)
	I will often participate in activities on short video social platforms	
	I often create videos and share them on short video social platforms	
	I often login on short video social platforms	
	I plan to extend my usage time of short video social platforms	

5. Data analysis and results

Structural equation modeling (SEM) was adopted to test the research hypotheses of the study. It is important to test the reliability and validity of the research variables. The study used SPSS 24.0 software to describe the population characteristics of the samples, reliability, and Cronbach's α through descriptive analysis. AMOS 23 was used to test the fit of the measurement model; Harman single-factor was used to test for CMB, and VIF was used to test for the collinearity of variables. Finally, the multi-group structural equation model was used to determine whether there are significant differences in the paths under different conditions of usage time grouping. This study found that Cronbach's α coefficients of positive consumer participation behavior, negative consumer participation behavior, consumer perceived value, consumer happiness, and consumer stickiness were 0.880, 0.862, 0.887, 0.920, and 0.894, respectively. All values were greater than 0.8, which indicates that each scale of the study had better reliability.

The Harman single-factor test was used to conduct the homology variance test on the recovered data (Table 3). All the measurement data in this paper were entered into the SPSS software for unrotated factor analysis. The variation degree caused by one factor was tested, and the result revealed that the variation explained by the first component was 34.787%, which was less than 40%, so the

Table 3. Collinearity diagnosis.

	Collinearity statistics	
	Tolerance	VIF
Gender	0.874	1.144
Age	0.850	1.177
Educational level	0.957	1.044
Usage time	0.958	1.044
Positive consumer participation behavior	0.821	1.218
Negative consumer participation behavior	0.844	1.184
Consumer perceived value	0.575	1.739
Consumer happiness	0.686	1.457
Consumer stickiness	0.531	1.883

problem of homology variance was not serious. The VIF of each variable was lower than 2, which indicates that no collinearity exists.

Validity determines whether the measurement results can truly reflect the actual situation. In this paper, content, convergence, and discriminant validity were tested. Table 4 shows that the CR of each variable is greater than 0.7, and AVE is greater than 0.5. From the discriminant validity, it can be seen that the AVE square of each variable is greater than the correlation coefficient between variables, indicating that it has better discriminant validity.

The fitting indexes of the SEM in this study were CFI = 0.987, NFI = 0.965, and RMSEA = 0.025, which met the standard values of modeling fit. Table 5 indicates that positive consumer participation behavior has a significant positive impact on consumer perceived value ($\beta = 0.399$, $P < 0.001$); thus, H1 holds. Negative consumer participation behavior has a significant negative impact on consumer perceived value ($\beta = -0.334$, $P < 0.001$); H2 holds. Consumer perceived value has a significant positive impact on consumer happiness ($\beta = 0.502$, $P < 0.001$); H4 holds. Finally, consumer perceived value has a significant positive impact on consumer stickiness ($\beta = 0.318$, $P < 0.001$); H6 holds.

The commonly used method of mediator effect test is causality, but its statistical efficacy is relatively low, so the Sobel test is often used as a supplement. However, the Sobel test also has drawbacks. It needs to assume that the sample distribution of indirect effects is normal, so this paper used the bootstrap method proposed by Hayes (2013). As suggested by Hayes, the bootstrap sample number should be set to 5000, and the mediator effect test should be conducted. The bootstrap confidence interval does not contain 0, and the corresponding indirect, direct, or total effect exists. From Table 6, it can be concluded that the indirect estimate of consumer perceived value-consumer happiness-consumer stickiness path is 0.160, 95% confidence interval [0.125, 0.202], excluding 0, indicating that the mediator effect is significant; thus, H6 holds. Meanwhile, the indirect estimate of consumer participation behaviors-consumer perceived value-citizenship behavior path is 0.200, 95% confidence interval [0.157, 0.251], excluding 0, indicating that the mediator effect is significant; the indirect estimate of negative consumer participation behavior-consumer perceived value-citizenship behavior path is −0.168, 95% confidence interval [−0.211, −0.126], excluding 0, indicating that the mediator effect is significant; the indirect estimate of positive consumer participation behavior-consumer perceived value-consumer stickiness path is 0.208, 95% confidence interval [0.165, 0.257], excluding 0, indicating that the mediator effect is significant; and the indirect estimate of negative consumer participation behavior-consumer perceived value-consumer stickiness path is −0.174, 95% confidence interval [−0.217, −0.132], including 0, indicating that the mediator effect is significant. From the above tests, it can be seen that the consumer perceived value plays a mediating role between different emotional consumer participation behaviors and consumer happiness or consumer stickiness.

To determine whether there is a significant difference in the impact of the model for different usage times, the usage time was set as less than 6 months as the low group and more than 6 months as the high group to perform multigroup SEM and to perform a multigroup comparison between the restricted and unrestricted models (presented in Table 7). Only the impact of positive

Table 4. Descriptive statistics, correlations, AVE.

	M	SD	AVE	Usage time	Positive consumer participation behavior	Negative consumer participation behavior	Consumer perceived value	Consumer happiness	Consumer stickiness
Usage time	0.482	0.500	—	—					
Positive consumer participation behavior	4.804	1.217	0.649	−0.018	0.806				
Negative consumer participation behavior	4.865	1.206	0.611	0.007	−0.229***	0.782			
Consumer perceived value	4.894	1.198	0.666	0.055	0.412***	−0.371***	0.816		
Consumer happiness	4.901	1.183	0.660	−0.029	0.161***	−0.216***	0.431***	0.812	
Consumer stickiness	4.907	1.146	0.630	0.032	0.208***	−0.253***	0.496***	0.496***	0.794

Notes: *** $P < 0.001$, SD = standard deviation.

Table 5. Descriptive statistics, correlation analysis, AVE.

	Consumer perceived value	Consumer happiness	Consumer stickiness
Control variables			
Gender			0.603***
Age			−0.334***
Educational level			−0.014
Main effects			
Positive consumer participation behavior	0.399***		
Negative consumer participation behavior	−0.334***		
Consumer perceived value		0.502***	0.363***
Consumer happiness			0.318***
R^2	0.314	0.218	0.542

Notes: *$P < 0.05$; ***$P < 0.001$. Parameters are unstandardized. The fit statistics are as follows: CFI = 0.987; NFI = 0.965; RMSEA = 0.025.

Table 6. Bootstrap mediation verification.

Path	Indirect estimate	Standard error	Bias-Corrected (95% CI)		
			Lower	Upper	*P*
positive consumer participation behavior – consumer perceived value – consumer happiness	0.200	0.024	0.157	0.251	0.000
negative consumer participation behavior – consumer perceived value – consumer happiness	−0.168	0.022	−0.211	−0.126	0.000
positive consumer participation behavior – consumer perceived value – consumer stickiness	0.208	0.024	0.165	0.257	0.000
negative consumer participation behavior – consumer perceived value – consumer stickiness	−0.174	0.022	−0.217	−0.132	0.000
consumer perceived value – consumer happiness – consumer stickiness	0.160	0.019	0.125	0.202	0.000

consumer participation on consumer perceived value is significantly different between the high and low usage times ($\blacktriangle x^2(1) = 33.493$, $P < 0.001$); the path coefficient of positive consumer participation to consumer perceived value in the low usage time is 0.211 ($P < 0.001$), and the path coefficient of positive consumer participation in the high usage time is 0.651 ($P < 0.001$), indicating that usage time will enhance the positive impact of positive consumer participation on consumer perceived value;

Table 7. Comparative analysis of multigroup structural equation models.

	Consumer perceived value		Consumer happiness		consumer stickiness		
	Low group (n = 470)	High group (n = 438)	Low group (n = 470)	High group (n = 438)	Low group (n = 470)	High group (n = 438)	$\blacktriangle x^2(1)$
Control variables							
Gender					0.554***	0.644***	0.617
Age					−0.339***	−0.345***	0.006
Educational level					−0.068	0.024	0.794
Main effects							
Positive consumer participation behavior	0.211***	0.651***					33.493***
Negative consumer participation behavior	−0.155***	−0.563***					34.124***
Consumer perceived value			0.590***	0.446***	0.437***	0.312***	3.196a 3.269b
Consumer happiness					0.352***	0.276***	1.687
R^2	0.113	0.620	0.222	0.222	0.552	0.543	

Notes: *$P < 0.05$; ***$P < 0.001$. Parameters are unstandardized. The fit statistics: CFI = 0.976; NFI = 0.934; RMSEA = 0.024, a$\blacktriangle x^2(1)$ for consumer perceived value–consumer happiness; b$\blacktriangle x^2(1)$ for consumer perceived value–consumer stickiness.

thus, H7 holds. The impact of negative consumer participation on consumer perceived value is significantly different between the high and low usage times ($\blacktriangle x^2(1) = 34.124$, $P < 0.001$); the path coefficient in the low usage time is −0.155 ($P < 0.001$), and the path coefficient in the high usage time is −0.563 ($P < 0.001$), indicating that usage time will enhance the negative impact of negative consumer participation on consumer perceived value; thus, H8 holds.

6. Conclusion and discussion

This study explored the relationship between ICT and the sharing economy and found that ICT plays an indispensable role in the sharing economy's development. ICT promotes various resource-sharing activities and helps social platforms innovate consumer participation and interaction behaviors. Simultaneously, this study was conducted from the perspective of value co-creation and found that there is a close internal relationship between consumer participation behaviors and consumer stickiness. Value creation of short-form video-sharing mobile social platforms mainly emerges from the interactions between consumers. In this study, consumer participation behaviors are divided into positive and negative behaviors, and it was found that consumer participation behaviors have a significant impact on consumer perceived value under different emotions. Positive consumer participation behavior promotes consumer perceived value, whereas negative consumer participation behavior reduces it, as demonstrated by Lloyd and Luk (2011). The study found that different usage times have a significant regulating effect on consumer participation behaviors: long, positive participation positively impacts consumers' perceived value, whereas long, negative participation negatively impacts consumers' perceived value. Research result shows that consumer happiness plays a significant intermediary role between consumer perceived value and consumer stickiness. Consumer perceived value improves consumer happiness, and, in turn, consumer happiness promotes consumer stickiness, which establishes the internal mechanism between consumer participation behaviors and consumer stickiness on short-form video-sharing mobile social platforms.

6.1. Theoretical implications

This study contributes to the value co-creation theory under the sharing economy and the existing literature on consumer participation behaviors and consumer stickiness in several ways.

First, the value co-creation theory is expanded and enriched under the sharing economy model. Although value co-creation has been emphasized and received numerous empirical analysis (Cova et al., 2011; Jaakkola & Hakanen, 2013; Pongsakornrungsilp & Schroeder, 2011), this paper found that the process of value co-creation also changes from the traditional single-level structure to the multi-level structure at different service stages and value co-creation can be seen in every process. The connection between producers and consumers of goods and services has become closer. Second, this paper reveals the internal relationship between ICT and the development of the sharing economy, that ICT gave birth to the sharing economy as a technical condition and has played a role in promoting the development of the sharing economy. This paper will make up for the deficiency of the linkage between ICT and the value co-creation theory in the sharing economy research by expanding the research on consumer value co-creation behavior, filling in the gaps in the field of ICT and the sharing economy, and providing a new perspective for the sustainable research of ICT and the sharing economy. Third, the effect of positive consumer participation behaviors on short-form video-sharing mobile social platforms is explored alongside the negative behaviors, which is seldom mentioned in previous literature. Consumer happiness and usage time are introduced into the study to discuss their importance toward consumer stickiness, further enriching the scope of theoretical research on consumer behavior and extending previous literature (e.g. Dai & Gu, 2017; Kim & Tang, 2020).

This paper concludes that consumer-side variables, such as perceived value and happiness, can affect stickiness, which is a platform-side variable. Through this research, we can have a clear

understanding of the internal mechanism of consumer participation behaviors and consumer stickiness on short-form video-sharing social media platforms. It expands the theoretical research between participation behaviors and stickiness and lays the theoretical foundation for follow-up research.

6.2. Practical implications

This research has several important implications for managers of social media companies. First, greater efforts are needed to ensure positive participation, the improvement of consumer perceived value, the SWB of consumers, and the increase of consumer stickiness. Managers should pay attention to the cultivation of consumer behavior and promote consumers' positive participation and avoid negative participation through rewards, interaction, or other new ways or features for participation remained to be activated. To better serve consumers' perceived value, platforms need to continuously optimize their algorithms so that they provide consumers with more precisely personalized content. Consumer perceived social value needs the establishment of a stable social group, and economic value needs the profits gaining in the process of uploading works and interactions. They all need a process of continuous accumulation and reinforcement with a long cycle, which requires the platform to pay attention to the long-term management of consumer relationships and the improvement of feedback mechanisms. Second, sharing economy enterprises should be committed to continuously promoting the value co-creation behavior of consumers. For short-form video-sharing mobile social platforms, some rewards can be provided to encourage common users' video creation intention and other value creation activities, such as advertising, selling goods, broadcasting news, etc. For enterprises in other fields, similar strategies could be adopted to secure better development. consumer participation and interaction behaviors in the short-form video-sharing mobile social platform are also realized based on ICT. Therefore, shared enterprises should continue to actively explore the role of ICT in the sharing economy to continuously improve service quality and innovate the platform. Moreover, ICT should be promoted to meet technological needs in various fields and contribute to the advancement of societies.

The sharing economy has adapted to the energy crisis and has a positive impact on social development, wherein more measures need to be adopted to increase this positive impact. First, the government should roll out policies to support the regulated development of diversified online social networking and short video platforms and encourage micro-innovation, micro-applications, micro-products, micro-movies, and other innovations. Second, the development of the sharing economy should become a bridge connecting the development of multiple fields, such as education and tourism development. Besides, the emergence of the sharing economy has also engendered new employment methods, such as Internet celebrities who advertise and sell goods online. In today's increasingly severe employment situation, this new way of employment is worth promoting and encouraging.

7. Limitations and future research

Due to the novelty of this paper's research direction, there is relatively limited literature about the internal mechanism of consumer value co-creation in the sharing economy. TikTok has only become a phenomenon-level application in the past three years. At present, research on consumer participation behaviors on a short-form video-sharing mobile social platform is not extensive enough. There is no clear definition of the consumer's perceived value and no uniform standard for the measurement items selected by different scholars.

In future research, we will focus more on the internal mechanism of value co-creation under the sharing economy model. Through the gradual deepening of the research on the value co-creation behavior among consumers, we will build a theoretical model of value co-creation under the sharing economy model and enrich the research on value co-creation theory. Simultaneously,

follow-up research can be extended to other business models of the sharing economy, and they can further study the internal factors of consumer value co-creation behavior. While consumer participation behaviors promote, value creation may also engender destruction. However, research on value destruction is still at the nascent stages and requires further study in future research. Users of short-video mobile platforms may get monetary incentives to create videos. However, consumers' participation behaviors may be influenced by the risk of privacy and security in the online environment. Future research may examine the moderating impacts of monetary gains and risks of privacy concerns on consumers' participation behaviors in online communication platforms by considering the relationships of the constructs presented in this study as a guide.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the Natural Science Foundation of China (grant number 71831005) and Shenzhen Humanities & Social Sciences Key Research Bases (grant number KP191001).

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